

# SOFTWARE ENGINEERING 381 - SEN381

## ASSIGNMENT 3(A3)

### Research for Quality, Security & Production Readiness Decisions

|                             |                                                                                                                                                  |
|-----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Assessment type</b>      | Team research assignment                                                                                                                         |
| <b>Team size</b>            | 3 students - same team used for the integrated project                                                                                           |
| <b>Duration</b>             | From opening in Week 3 to the stated Week 4 submission deadline                                                                                  |
| <b>Total marks</b>          | 50                                                                                                                                               |
| <b>Submission</b>           | One concise professional engineering research brief per team<br><br>Include a cover and table of contents – Explicitly mention nonparticipation. |
| <b>Project relationship</b> | Research preparation for Milestone 3; CivicConnect Milestone 3(M3) decisions and evidence are NOT produced in this assignment                    |
| <b>Referencing</b>          | Institution-prescribed academic referencing style                                                                                                |
| <b>Academic year</b>        | 2026                                                                                                                                             |

#### Central principle

A3 researches → Milestone 3(M3) decides, applies, verifies and provides evidence. Research should expand the team's engineering options and judgement; it must not prematurely manufacture CivicConnect quality, security, deployment or operational decisions.

## 1. Purpose of the Assignment

Assignment 3 prepares students for the quality, security, deployment and operational decisions that become increasingly important as a software product approaches a more complete and defensible engineering state. The assignment draws forward selected Week 5 and Week 6 concerns so that students research credible approaches before they are required to make and defend Milestone 3 decisions.

- Research how software quality should be planned and verified according to risk rather than treated as testing at the end.
- Research how security threats and requirements can be translated into practical engineering controls and verification evidence.
- Research what makes software production-ready, deployable, recoverable and observable rather than merely functional in a development environment.

- Compare viable approaches, tools or techniques and identify their limitations and trade-offs.
- Create a clear research-to-M3 bridge without making CivicConnect M3 decisions prematurely.

#### Important boundary

Assignment 3 may inform Milestone 3, but it is not Milestone 3. Students must NOT present researched approaches as already-selected CivicConnect controls, test strategy, security design, deployment platform, monitoring solution or release decision. Those decisions must later be made against CivicConnect's actual requirements, architecture, implementation state, risks, constraints and available evidence.

## 1.1 SEN381 Progression

Assignments 1 and 2 established the habit of researching before committing to engineering decisions. Assignment 3 continues that progression at a later lifecycle point. Students are now expected to connect research to measurable evidence, product risk and operational consequences rather than produce descriptive summaries.

#### Progression

A1: research foundations for engineering decisions → A2: research design, persistence, API and collaborative-engineering options → A3: research quality, security and production-readiness options → M3: decide, apply and defend appropriate controls using actual project evidence.

## 2. Learning Focus

This assignment supports the Week 5 and Week 6 emphasis on Quality Engineering, security throughout development, production readiness, deployment and operations. It also reinforces the Week 4 principle that engineering evidence should be controlled, traceable and suitable for defensible progression decisions.

## 3. Submission Expectations

- Submit one team document answering all four questions in the order given.
- Use concise professional writing. Tables, diagrams, evidence maps and comparison models are encouraged where they communicate analysis more effectively than prose.
- Use at least 8 credible sources across the assignment. A stronger submission normally combines academic/standards sources, recognised professional sources and official tool/platform documentation where relevant.
- Use current evidence for contemporary security, testing, deployment, cloud/platform and observability claims.
- Do not use product marketing claims as independent evidence of quality, security, scalability, reliability or suitability.
- AI tools are not academic sources. Important AI-assisted claims must be independently verified.
- Include a reference list and the AI Research & Verification Record in Section 9.
- Each team member must make a meaningful contribution and must understand the complete submission.
- Quality of evidence, comparison and engineering reasoning is valued over page count.

## 4. Mark Allocation

| Question | Focus                                                   | Marks |
|----------|---------------------------------------------------------|-------|
| Q1       | Quality Strategy & Risk-Based Verification              | 15    |
| Q2       | Security Engineering & Threat-Based Controls            | 15    |
| Q3       | Production Readiness, Deployment & Operational Evidence | 15    |
| Q4       | Research-to-M3 Engineering Map                          | 5     |
|          | TOTAL                                                   | 50    |

## 5. Question 1 - Quality Strategy & Risk-Based Verification [15]

Research how a software team can establish a quality strategy in which verification effort is driven by product risk, quality expectations and the consequences of failure rather than by attempting to test everything equally.

Your response must address the following:

- Explain the relationship between Quality Assurance (QA), Quality Control (QC), Verification and Validation (V&V) only to the extent needed to show how they contribute to an overall quality strategy.
- Research risk-based testing/verification and explain how likelihood, impact and business/technical criticality can influence what receives deeper verification.
- Compare at least THREE complementary forms of verification evidence appropriate to a modern web-based information system. These may include unit, integration/API, system/end-to-end, static analysis, regression, performance/load or other justified evidence.
- For each selected form of evidence, explain what type of failure or quality risk it is strong at exposing and what it cannot prove on its own.
- Research how automation and quality gates can make verification repeatable while recognising the danger of treating a green pipeline as proof of complete quality.

Required visual: Risk-to-Verification Evidence Map

Create ONE original visual that traces at least FOUR example software risks or quality concerns to appropriate verification evidence and a progression/quality decision. The visual must show reasoning, not merely list testing types.

### Critical question

Why is 'all automated tests passed' insufficient evidence, by itself, to conclude that a software product is high quality or ready for release?

| Criterion                                                         | Marks |
|-------------------------------------------------------------------|-------|
| Understanding of quality strategy and risk-based verification     | 4     |
| Comparison of complementary verification evidence and limitations | 4     |

|                                                           |   |
|-----------------------------------------------------------|---|
| Risk-to-Verification Evidence Map: accuracy and reasoning | 3 |
| Automation/quality-gate analysis and critical judgement   | 2 |
| Research evidence and synthesis                           | 2 |

## 6. Question 2 - Security Engineering & Threat-Based Controls [15]

Research how security should be engineered throughout software development rather than added as a final penetration-testing activity. The purpose is to understand how credible threats lead to security requirements, design/construction controls and verification evidence.

Required response:

- Explain the purpose of threat modelling and research ONE recognised threat-modelling approach or structured method suitable for software engineering. Focus on how it supports decisions rather than reproducing a catalogue of threat terms.
- Select FOUR security concern areas relevant to a modern web-based information system. At least THREE of the following must be represented: authentication, authorisation/access control, secrets/configuration, dependency vulnerabilities, input/data handling, sensitive information/privacy, logging/auditability.
- For each selected concern, show the chain: plausible threat or misuse → security requirement/control objective → engineering control → verification evidence.
- Explain how secure design, secure coding and dependency/security checks complement one another rather than acting as substitutes.
- Evaluate at least TWO limitations of relying only on automated security tools or AI-generated security recommendations.

Required model: Threat-to-Control Traceability Table

| Threat / misuse                          | Security requirement or control objective | Possible engineering control              | Verification evidence                  | Residual risk / limitation |
|------------------------------------------|-------------------------------------------|-------------------------------------------|----------------------------------------|----------------------------|
| Example only - students replace this row | What must be protected or prevented?      | How might the requirement be implemented? | What evidence could support the claim? | What remains uncertain?    |

### Critical question

If a security scanner reports no high-severity findings, what important security claims may still remain unproven?

| Criterion                                                               | Marks |
|-------------------------------------------------------------------------|-------|
| Threat-modelling research and engineering relevance                     | 3     |
| Four security concern chains: threat → requirement → control → evidence | 5     |

|                                                               |   |
|---------------------------------------------------------------|---|
| Relationship between secure design, construction and checking | 2 |
| Critical analysis of automated/AI security limitations        | 3 |
| Research evidence and synthesis                               | 2 |

## 7. Question 3 - Production Readiness, Deployment & Operational Evidence [15]

Research what must be considered before a software product can be described as production-ready. The response must move beyond 'deploy the application to the cloud' and investigate the engineering evidence needed to deploy, recover, operate and observe a system safely.

Research and analyse the following areas:

- Environment strategy: development, test/staging and production; configuration and environment parity; why uncontrolled environment differences create deployment risk.
- Secrets and production configuration: how sensitive configuration should be separated from source code and controlled appropriately.
- Release and deployment: automated/repeatable deployment, release controls and the difference between successfully building software and successfully deploying a usable product.
- Failure and recovery: rollback, database/data compatibility, backup/recovery considerations and what should happen when a release fails.
- Operations and observability: logs, metrics, monitoring and alerts; explain what each can contribute to operational evidence and why collecting data without actionable interpretation is insufficient.
- Operational quality: investigate how reliability, performance, scalability, cost and supportability can affect production-readiness decisions.

Required comparison: Production-Readiness Evidence Matrix

Create ONE matrix containing at least SIX production-readiness concerns. For each concern identify: the risk if ignored, evidence that should exist before release, and evidence that should continue to be observed after release.

### Critical question

Why can software that passes functional tests and runs successfully on a developer's machine still be unready for production?

| Criterion                                           | Marks |
|-----------------------------------------------------|-------|
| Environment, configuration and deployment research  | 4     |
| Failure/recovery and operational-readiness analysis | 3     |
| Observability and operational-quality reasoning     | 3     |
| Production-Readiness Evidence Matrix                | 3     |

## 8. Question 4 - Research-to-M3 Engineering Map [5]

Convert the strongest findings from Questions 1-3 into a concise bridge for later Milestone 3 decision-making. This is NOT a CivicConnect solution and must not state that a specific tool, test suite, security control, deployment platform or monitoring approach has already been selected.

Create a Research-to-M3 Engineering Map containing at least SIX meaningful findings across quality, security and production readiness.

| Research finding                                                            | Engineering concern it addresses   | Candidate approach / evidence to consider | Trade-off / limitation found in research                               | Decision M3 must still make                                                      |
|-----------------------------------------------------------------------------|------------------------------------|-------------------------------------------|------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| Example: API integration tests can expose contract/integration regressions. | Interface/integration failure risk | Automated API/integration verification    | Requires representative environment/data and does not prove usability. | Which CivicConnect interactions are sufficiently risky to require this evidence? |

The final column is essential. It demonstrates that research informs the project without pretending that research has already made the project decision.

| Criterion                                                        | Marks |
|------------------------------------------------------------------|-------|
| Six relevant findings spanning Q1-Q3                             | 2     |
| Clear transfer from research to candidate M3 evidence/approaches | 2     |
| Preservation of the research-versus-project-decision boundary    | 1     |

## 9. Research, Referencing and AI Evidence Standard

### 9.1 Research quality

- Use sources that are credible for the claim being made. Standards, academic research and recognised professional engineering guidance are preferable for general principles; official documentation is appropriate for specific tool/platform capabilities.
- Distinguish researched fact from team interpretation or recommendation.
- Where research sources disagree or an approach is context-dependent, acknowledge the limitation rather than forcing a universal conclusion.
- Paraphrase and synthesise. Do not construct the assignment from long quotations.
- Apply the institution-prescribed referencing style consistently in-text and in the final reference list.

- Fabricated sources, unverifiable citations or unsupported technical claims undermine the relevant criterion.

## 9.2 AI Research & Verification Record

If AI is used, include a short appendix using the structure below. The record is not awarded marks simply for existing; poor verification can reduce the quality of the relevant answer.

| Team member | AI tool / use | Purpose | Output used? | How independently verified | What was changed/rejected |
|-------------|---------------|---------|--------------|----------------------------|---------------------------|
|             |               |         |              |                            |                           |
|             |               |         |              |                            |                           |

## 9.3 Progressive Engineering Evidence

By Assignment 3 the team is already working within the controlled project environment. Research and document development should therefore leave authentic, progressive evidence rather than appearing only as a completed upload.

- Retain meaningful research and document-development history in the team repository.
- Use meaningful commits and preserve progressive revision rather than reconstructing activity immediately before submission.
- Where an assignment artefact is reviewed through a Pull Request, preserve the normal project governance and accountability expectations.
- Do not mix unapproved M3 implementation decisions into Assignment 3 merely to make the research appear more project-specific.

## 10. Team Accountability

The assignment is completed in the same teams of three used for the integrated project. The submission is shared, but all members remain responsible for its integrity and must understand the complete work.

- Dividing questions among team members does not remove collective responsibility for the final submission.
- The team should retain reasonable evidence of research, contribution, review and document revision.
- Lecturers may ask brief verification questions where contribution, source use or understanding is unclear.
- AI assistance does not transfer accountability away from the student team.

## 11. What Assignment 3 Must NOT Become

| Do not                                                               | Instead                                                                                                           |
|----------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| Write a textbook chapter describing every Week 5 and Week 6 concept. | Select and analyse evidence, relationships, limitations and consequences relevant to later engineering decisions. |
| Create the CivicConnect M3 quality strategy or test plan.            | Research how quality strategy and risk-based verification should be designed.                                     |

|                                                             |                                                                                                          |
|-------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| Declare CivicConnect security controls as already selected. | Research threat-to-control reasoning and the evidence different controls can provide.                    |
| Choose the CivicConnect hosting/deployment platform.        | Research production-readiness, deployment and operational evidence that should influence a later choice. |
| Treat a green pipeline, scanner or AI answer as proof.      | Explain what the evidence supports, what it does not prove and what complementary evidence is needed.    |
| Recommend tools because they are popular or familiar.       | Use credible evidence, project-relevant criteria and explicit trade-offs.                                |
| Turn Q4 into a hidden M3 solution.                          | State what M3 must still decide using CivicConnect's actual context.                                     |

## 12. Submission Checklist

- ☐ One team document submitted and all four questions answered in order.
- ☐ Q1 includes a Risk-to-Verification Evidence Map.
- ☐ Q2 includes four threat/control/evidence chains and the Threat-to-Control Traceability Table.
- ☐ Q3 includes a Production-Readiness Evidence Matrix with at least six concerns.
- ☐ Q4 includes at least six Research-to-M3 findings across quality, security and production readiness.
- ☐ At least 8 credible sources are used across the assignment.
- ☐ Important contemporary technical claims are supported by appropriate current evidence.
- ☐ In-text citations and the reference list are consistent.
- ☐ AI Research & Verification Record is included if AI was used.
- ☐ No CivicConnect M3 quality, security, deployment or operational decision has been made prematurely.
- ☐ Progressive team contribution/research evidence has been retained.

### Assignment 3 takeaway

Research should not tell the team what to copy into Milestone 3. It should give the team enough evidence to ask better questions, compare viable controls and decide what quality, security and production-readiness evidence CivicConnect actually needs.

**Academic Integrity and Referencing Disclaimer:** All submitted work must include appropriate in-text citations and a corresponding reference list. Failure to provide in-text citations and references will result in an automatic mark of zero (0). Plagiarism, fabricated or misrepresented research/evidence, and inappropriate presentation of another person's or AI-generated work as the team's own are subject to the applicable academic-integrity requirements.